

Revised Generalized RISEI Theory

Protocol-Resolved Interventional Response, Protected Riesz Geometry,
and Limit-Ordered Functional Hierarchies

21 July 2026

Abstract

We revise the generalized RISEI framework without discarding its protocol-resolved response hierarchy. The original generalized theory treated finite-time, multi-time, finite-resource, and ordered interventions within finite-dimensional time-local GKSL dynamics. The present revision adds the spectral and geometric structures required by the strongest currently supported RISEI claim: response-relevant protected Riesz subspaces, cluster-resolved spectral flow, source/readout selection maps, punctured selection manifolds, admissible perturbation classes, and an explicit ordering of strong-dissipation, kernel-lifting, continuum, and infinite-volume limits. The central phenomenon is no longer stated as an unconditional property of arbitrary GKSL systems. It is formulated as a kernel-conditioned response law: under a semisimple response-relevant protected subspace, nonvanishing Schur–Zeno coupling, regular source/readout selection, and declared observability conditions, a nonadditive Möbius response can retain finite peak height while its integrated weight vanishes. The local codimension of the corresponding observable-selection set is the real Jacobian rank of the active constraint map, not universally one. Exact kernels and approximate kernels are distinguished: fixed kernel-lifting perturbations generally remove the asymptotic hierarchy, while double-scaling limits may retain a finite-window crossover. Definitions, exact statements, conditional theorems, conjectures, model-specific numerical laws, and certification diagnostics are explicitly separated.

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1 Scope and organizing principle

The revised theory retains the operational content of generalized RISEI:

1. physical intervention on identified sectors;
2. comparison of full and intervened responses;
3. subset-Möbius and order-resolved irreducible decomposition;
4. finite-time and multi-time response hierarchies;
5. functional-dependent asymptotic and resource classification.

It adds a sixth structural layer:

6. response-relevant protected Riesz geometry and its observable-selection map.

The central conceptual replacement is

protocol-resolved sector response hierarchy $\implies$ protocol-resolved response hierarchy + protected Riesz geometry + declared limit protocol.

(1)

The previous generalized theory is recovered by omitting the additional spectral-geometric data. The original stationary scalar RISEI theory is further recovered as the stationary, linear, static, single-intervention section.

2 Logical status and claim discipline

Every statement in this document belongs to one of the following levels.

Definition.

An object or classification declared by the framework.

Exact. A theorem proved from explicitly stated assumptions.

Conditional.

An exact implication under additional spectral, geometric, or observational hypotheses.

Conjecture.

A proposed general law supported by partial analysis or multiple model classes but not yet proved.

Numerical phenomenon.

A robust observation in a specified physical or random GKSL family.

A finite ensemble is never identified with a universality proof. A model-specific coefficient is never promoted to a universal constant merely because it has many decimal places.

3 Dynamical and operational assumptions

Assumption 3.1 (Finite-dimensional time-local GKSL dynamics). The primary theory is formulated on a finite-dimensional Liouville space. For every admissible intervention protocol π and probe h ,

$$\frac{d\rho}{dt} = \mathcal{L}_{\Gamma,\pi,h}(t)\rho(t), \quad (2)$$

where the instantaneous generator is of GKSL form. Non-Markovian memory kernels, initial system–environment correlations, and continuum operators with essential spectrum are deferred extensions.

Assumption 3.2 (Physical admissibility of interventions). An intervention is represented by a physical generator modification,

$$\mathcal{L}_{\Gamma,\pi,h}(t) = \mathcal{L}_0(\Gamma, t) + \sum_{j=1}^m \kappa_j(t) \mathcal{G}_{S_j}(t) + \mathcal{V}_h(t), \quad (3)$$

for which the total instantaneous generator remains GKSL admissible. Formal deletion of arbitrary matrix entries is not a primary definition.

Definition 3.3 (Intervention protocol). A protocol is the ordered finite-resource object

$$\pi = [(S_1, \kappa_1, I_1), \dots, (S_m, \kappa_m, I_m)], \quad (4)$$

where S_j is an operational sector, κ_j is its strength profile, and I_j is its active time window. Protocols with the same sector set but different order, envelope, or duration are distinct.

Assumption 3.4 (Finite-time primacy). The primary dynamical object is

$$\mathcal{U}_\pi(t, t_0) = \mathcal{T} \exp \left[\int_{t_0}^t \mathcal{L}_{\Gamma,\pi,0}(\tau) d\tau \right]. \quad (5)$$

Stationary and frequency-domain responses are derived limits or transforms.

Assumption 3.5 (Fixed comparison class). Unless promoted to explicit protocol data, the initial state, probe, readout, observation window, operator ordering, measurement scheme, and normalization are held fixed when different cuts or protocols are compared.

Assumption 3.6 (Regular asymptotic representation). For a declared functional Φ and limit path, the response may have the form

$$\Phi[R](\Gamma) = \Gamma^{-\nu}(\log \Gamma)^m(C + o(1)), \quad C \neq 0, \quad (6)$$

with real or fractional ν . When no single regular variation exists, upper/lower valuations or piecewise classes are used.

Definition 3.7 (Admissible perturbation class). An admissible class $\mathcal{A}$ is a declared set of generators, sources, readouts, and protocol perturbations for which the hypotheses of a stated RISEI result remain valid. It must specify which perturbations preserve the protected spectral subspace, which merely displace the selection set, and which lift or destroy the protected subspace.

Definition 3.8 (Limit protocol). A limit protocol Λ is an ordered list of limits and fixed scaling combinations, for example

$$\Lambda = (\Gamma \rightarrow \infty; \varepsilon \rightarrow 0; N \rightarrow \infty; L \rightarrow \infty), \quad (7)$$

possibly with a double-scaling variable such as $x = \Gamma \lambda_{\min}(\varepsilon, N, L)/J_{\text{eff}}$ held fixed. Two results obtained with different limit orderings are not silently identified.

4 Protocol-resolved response hierarchy

For a fixed measurement convention, let

$$C_{\pi}^{(n)}(t_n, \dots, t_1) = \langle O_n(t_n) \cdots O_1(t_1) \rangle_{\pi, c} \quad (8)$$

denote the connected n -time cumulant. The ordering and measurement backaction belong to the observational protocol.

Definition 4.1 (Interventional response). Let $\emptyset$ be the no-intervention protocol. The order- n response is

$$R_{\pi}^{(n)}(t_n, \dots, t_1) = C_{\emptyset}^{(n)}(t_n, \dots, t_1) - C_{\pi}^{(n)}(t_n, \dots, t_1). \quad (9)$$

The sign convention is full minus intervention.

When a counting process is available, a tilted generator defines

$$Z_{\pi}(\lambda, t) = \text{Tr}[\mathcal{U}_{\pi, \lambda}(t, 0)\rho_0], \quad (10)$$

$$K_{\pi}^{(n)}(t) = \partial_{\lambda}^n \log Z_{\pi}(\lambda, t)|_{\lambda=0}. \quad (11)$$

Thus means, noise, higher cumulants, and full counting statistics can be compared within the same intervention language.

5 Functionals, valuations, and the reverse functional hierarchy

Definition 5.1 (Response functional). A response functional maps a response to a nonnegative feature. Important examples are

$$\Phi_{\text{pt}}[R] = |R(t_*, \omega_*)|, \quad (12)$$

$$\Phi_{\text{max}}[R] = \sup_{t \in I} |R(t)|, \quad (13)$$

$$N_{\infty}[R] = \sup_{\omega \in W} |R(\omega)|, \quad (14)$$

$$N_1[R] = \int_W |R(\omega)| d\omega, \quad (15)$$

$$\Phi_{\text{sgn}}[R] = \left| \text{P. V.} \int_W R(\omega) d\omega \right|, \quad (16)$$

$$\Phi_T[R] = \int_0^T |R(t)| dt. \quad (17)$$

The window W and all normalization conventions are part of the functional definition.

Definition 5.2 (Asymptotic valuation). For nonnegative $\Phi[R](\Gamma)$,

$$\nu_\Phi[R] = - \lim_{\Gamma \rightarrow \infty} \frac{\log \Phi[R](\Gamma)}{\log \Gamma}, \quad (18)$$

when the limit exists.

Definition 5.3 (Scaling signature). A finite signature is a declared tuple

$$\Sigma_\pi^{(n)} = (\nu_\infty, \nu_1, \nu_{\text{pt}}, z_t, z_\omega, m_{\log}, \dots). \quad (19)$$

Only components needed for the physical claim are required.

6 Irreducible sector structure

Fix a finite sector set T and a common intervention convention for every subset $U \subseteq T$.

Definition 6.1 (Subset-Möbius irreducible response).

$$\Omega_T^{(n)} = \sum_{U \subseteq T} (-1)^{|T|-|U|} C_U^{(n)}. \quad (20)$$

Theorem 6.2 (Uniqueness of the subset decomposition). *The family $\{\Omega_T^{(n)}\}$ is the unique family satisfying*

$$C_U^{(n)} = \sum_{T \subseteq U} \Omega_T^{(n)} \quad (21)$$

for every U .

Proof. This is finite Möbius inversion on the Boolean subset lattice. $\square$

Definition 6.3 (Functional-dependent protection depth).

$$k_{\min}^{(n, \Phi)} = \min\{|T| : \nu_\Phi[\Omega_T^{(n)}] = 0\}, \quad (22)$$

with $k_{\min}^{(n, \Phi)} = \infty$ when the set is empty.

Definition 6.4 (Two-sector Pattern (b)). For sectors 1, 2, the stationary first-order response has Pattern (b) when

$$\nu_\infty[\Omega_1] > 0, \quad \nu_\infty[\Omega_2] > 0, \quad \nu_\infty[\Omega_{12}] = 0, \quad (23)$$

$$\nu_1[\Omega_1] > 0, \quad \nu_1[\Omega_2] > 0, \quad \nu_1[\Omega_{12}] > 0. \quad (24)$$

Equivalently,

$$(k_{\min}^{(1, \infty)}, k_{\min}^{(1, 1)}) = (2, \infty). \quad (25)$$

The finite height of the joint response and the vanishing of its integrated weight are logically distinct requirements.

Definition 6.5 (Order-irreducible response).

$$\Xi_{S, T}^{(n)} = R_{S \rightarrow T}^{(n)} - R_{T \rightarrow S}^{(n)}. \quad (26)$$

A nonzero value witnesses order dependence, not by itself contextuality.

7 Protected Riesz subspaces and cluster-resolved response

Consider a time-independent generator for which a spectral subset $\mathcal{C}$ is separated by a closed contour $\gamma_{\mathcal{C}}$.

Definition 7.1 (Riesz projector and spectral cluster).

$$P_{\mathcal{C}}(\Gamma) = \frac{1}{2\pi i} \oint_{\gamma_{\mathcal{C}}} (z - \mathcal{L}_{\Gamma})^{-1} dz. \quad (27)$$

The range of $P_{\mathcal{C}}$ is the spectral cluster subspace. The projector, rather than an arbitrary ordering of individual poles, is the invariant object.

Proposition 7.2 (Cluster invariance). *As long as the contour remains in the resolvent set, $P_{\mathcal{C}}$ is invariant under relabeling and permutation of eigenvalues inside $\gamma_{\mathcal{C}}$. Continuous pole exchange or an internal degeneracy therefore does not alter the cluster response.*

Definition 7.3 (Response-relevant protected subspace). Let p be the source vector and c the readout covector in Liouville space. A protected cluster projector P is response relevant for a declared family if at least one admissible protocol has

$$Pp \neq 0, \quad cP \neq 0, \quad (28)$$

and the corresponding cluster contributes to at least one declared Möbius response after all cut combinations are formed.

Definition 7.4 (Cluster response). For a cluster $\mathcal{C}$, define

$$\chi_{\mathcal{C}}(z) = cP_{\mathcal{C}}(z - \mathcal{L})^{-1}P_{\mathcal{C}}p. \quad (29)$$

For a Möbius component, cluster responses are combined only after the same contour/continuation convention is applied to all cuts.

Remark 7.5. Individual pole labels are useful numerical bookkeeping but are not theory data. The theory data are Riesz projectors, cluster residues, and cluster-level Möbius combinations.

8 Strong-dissipation Schur–Zeno structure

Assume a stationary strong-dissipation family

$$\mathcal{L}_{\Gamma} = \Gamma \mathcal{D} + \mathcal{B}. \quad (30)$$

Let P be the spectral projector onto a semisimple zero cluster of $\mathcal{D}$, and $Q = \mathbb{I} - P$.

Assumption 8.1 (Gapped semisimple protected cluster). The eigenvalue 0 of $\mathcal{D}$ is semisimple on $\text{Ran } P$, and $Q\mathcal{D}Q$ is invertible with a declared spectral or resolvent gap.

Theorem 8.2 (Schur–Zeno expansion). *Under Theorem 8.1, the effective slow generator is*

$$S_{\Gamma}(z) = P(\mathcal{L}_{\Gamma} - z)P - P\mathcal{L}_{\Gamma}Q[Q(\mathcal{L}_{\Gamma} - z)Q]^{-1}Q\mathcal{L}_{\Gamma}P \quad (31)$$

$$= P(\mathcal{B} - z)P - \Gamma^{-1}P\mathcal{B}Q(Q\mathcal{D}Q)^{-1}Q\mathcal{B}P + O(\Gamma^{-2}) \quad (32)$$

for z in a fixed compact slow spectral region, with a bound determined by the fast resolvent and $\mathcal{B}$.

Proof. Factor $Q(\mathcal{L}_{\Gamma} - z)Q = \Gamma Q\mathcal{D}Q[\mathbb{I} + \Gamma^{-1}(Q\mathcal{D}Q)^{-1}Q(\mathcal{B} - z)Q]$ and expand the inverse by a convergent Neumann series for sufficiently large Γ . Substitution into the Schur complement gives the result. $\square$

Definition 8.3 (Schur–Zeno coupling). The leading slow correction is

$$K_Z = -PBQ(QDQ)^{-1}QB P. \quad (33)$$

A response-relevant Zeno linewidth requires a nonzero dissipative contribution of K_Z on the active protected cluster.

Proposition 8.4 (No protected cluster, no asymptotic narrow protected response). *If $\mathcal{D}$ is invertible on the entire response-relevant subspace and its inverse is uniformly bounded, then for fixed z in a compact set*

$$(\Gamma\mathcal{D} + \mathcal{B} - z)^{-1} = O(\Gamma^{-1}). \quad (34)$$

Consequently every bounded source/readout response on that subspace is $O(\Gamma^{-1})$ unless another singular scaling is introduced.

9 Source/readout selection geometry

Let $\theta \in \Theta$ denote observable, preparation, and calibration parameters. Define the projected source and readout data

$$s_P(\theta) = Pp(\theta), \quad r_P(\theta) = c(\theta)P. \quad (35)$$

The leading Möbius coefficients are generally bilinear functions of these data and of the effective cluster dynamics.

Definition 9.1 (Selection constraint map). A selection constraint map is a smooth real map

$$F : \Theta \longrightarrow \mathbb{R}^r \quad (36)$$

whose vanishing removes the leading unwanted protected overlap or leakage coefficient required by a declared functional hierarchy. Complex constraints are realified by separating real and imaginary parts.

Definition 9.2 (Observable-selection set and puncture). The raw selection set is $F^{-1}(0)$. The operational selection manifold is

$$\boxed{\mathcal{M} = F^{-1}(0) \setminus \mathcal{P}}, \quad (37)$$

where the puncture set $\mathcal{P}$ contains points at which the complementary source/readout overlap also vanishes, the joint cluster residue is zero, stability is lost, or the declared observability gate fails.

Theorem 9.3 (Rank–codimension theorem). *If $DF(\theta_0)$ has constant real rank r in a neighborhood of $\theta_0 \in F^{-1}(0)$, then $F^{-1}(0)$ is locally a smooth embedded submanifold of real codimension r .*

Proof. This is the constant-rank theorem, with the regular-value theorem as the full-rank special case. $\square$

Corollary 9.4. *Codimension one is not universal. A generic complex scalar constraint has real codimension two, while a symmetry-restricted real slice may reduce the rank to one.*

9.1 Two reciprocal branches in Candidate A

In the current minimal realization, the active real-slice constraint factorizes as

$$F(\theta_p, \theta_c) = f_p(\theta_p)f_c(\theta_c) = \cos \theta_p \cos \theta_c. \quad (38)$$

Hence

$$F^{-1}(0) = \left\{ \theta_p = \frac{\pi}{2} \right\} \cup \left\{ \theta_c = \frac{\pi}{2} \right\} \quad (39)$$

modulo periodic copies. Away from the intersection, the active constraint has rank one. At the intersection both reciprocal selection channels vanish, and the point belongs to the puncture set rather than to the Pattern (b) manifold. This formula is a model realization, not the universal definition of RISEI selection geometry.

10 Kernel-conditioned Pattern (b): central theorem schema

The strongest defensible RISEI claim is conditional. It is useful to separate an exact implication from the remaining analytic target.

Assumption 10.1 (Central structural hypotheses). For each model in a declared class:

- H1.** $\mathcal{L}_\Gamma = \Gamma\mathcal{D} + \mathcal{B}$ is a finite-dimensional stable GKSL family.
- H2.** $\mathcal{D}$ has a semisimple response-relevant protected Riesz subspace P .
- H3.** K_Z produces a nonzero slow linewidth of order J_{eff}^2/Γ .
- H4.** The leading single- and joint-sector cluster coefficients admit a smooth source/readout factorization with an active constraint map F of constant rank away from a puncture set.
- H5.** The relevant cluster residue is nonzero on $\mathcal{M}$, and the full Möbius combination passes declared cancellation and numerical-floor conditions.
- H6.** Perturbations are restricted to an admissible class that does not lift the protected cluster at an order dominating the Schur–Zeno linewidth.
- H7.** The limit protocol Λ is fixed.

Proposition 10.2 (Conditional functional hierarchy). *Assume [Theorem 10.1](#). Suppose further that on $\mathcal{M}$*

$$N_\infty[\Omega_1], N_\infty[\Omega_2] = O(\Gamma^{-a}), \quad a > 0, \quad (40)$$

$$N_\infty[\Omega_{12}] = C_\infty + o(1), \quad C_\infty > 0, \quad (41)$$

$$N_1[\Omega_1], N_1[\Omega_2], N_1[\Omega_{12}] = o(1). \quad (42)$$

Then Pattern (b) holds on $\mathcal{M}$, and its local codimension equals $\text{rank}_\mathbb{R} DF$.

Proof. The norm assumptions imply the valuations in [Theorem 6.4](#). The codimension statement follows from [Theorem 9.3](#); removal of $\mathcal{P}$ enforces nonzero operational visibility. $\square$

Conjecture 10.3 (General norm-hierarchy lemma). Under quantitative gap, residue, and cancellation bounds, the Schur–Zeno mechanism implies

$$N_\infty[\Omega_{12}] = O(1), \quad (43)$$

while

$$N_1[\Omega_{12}] = \begin{cases} O(\Gamma^{-1}), & \text{on the exact selection manifold,} \\ O((\log \Gamma)/\Gamma), & \text{in a scaled tube } u = \Gamma\delta/J_{\text{eff}} = O(1). \end{cases} \quad (44)$$

The logarithm is not assumed universal outside the rational cluster class for which it has been derived and numerically validated.

Conjecture 10.4 (Quantitative onset). There exists a model-dependent onset Γ_0 controlled by the fast resolvent bound, Schur linewidth, sector matrix elements, cluster residues, and cut-to-cut cancellation. A uniform Γ_0 over a class requires explicit lower and upper bounds on all of these quantities.

11 Finite- Γ calibration tubes

Let δ be a normal displacement from $\mathcal{M}$. The natural scaling variable is

$$u = \frac{\Gamma\delta}{J_{\text{eff}}}. \quad (45)$$

Definition 11.1 (Operational tube). For a declared response change $0 < q < 1$, define the finite- Γ tube radius $\delta_q(\Gamma)$ by

$$\frac{|N_\infty(\delta_q) - N_\infty(0)|}{N_\infty(0)} = q, \quad (46)$$

with tangent and normal directions specified by the singular vectors of DF .

Conjecture 11.2 (Tube collapse). Under the central structural hypotheses and a nondegenerate normal derivative,

$$\delta_q(\Gamma) = \frac{J_{\text{eff}}}{\Gamma} u_q + o(\Gamma^{-1}). \quad (47)$$

Thus a finite-width numerical band at fixed Γ collapses to the codimension- r selection manifold in the strong-dissipation limit; it is not an open asymptotic phase.

12 Exact kernels, approximate kernels, and noncommuting limits

Let a kernel-lifting perturbation produce

$$\mathcal{D}_\varepsilon = \mathcal{D}_0 + \varepsilon \mathcal{K}_{\text{tail}}, \quad (48)$$

where $\mathcal{D}_0$ has a protected kernel and the response-relevant minimum loss of $\mathcal{D}_\varepsilon$ is $\lambda_{\min}(\varepsilon) > 0$ for fixed $\varepsilon > 0$.

Proposition 12.1 (Fixed lifted kernel). *For fixed $\varepsilon > 0$, if $\mathcal{D}_\varepsilon$ is invertible on the response-relevant subspace with bounded inverse, the protected-kernel asymptotic mechanism is absent and [Theorem 8.4](#) applies as $\Gamma \rightarrow \infty$.*

Definition 12.2 (Kernel crossover variable).

$$x = \frac{\Gamma \lambda_{\min}(\varepsilon, N, L)}{J_{\text{eff}}}. \quad (49)$$

Here N and L may encode a grid and domain size in a continuum approximation.

Conjecture 12.3 (Exact-to-approximate protection crossover). For $x \ll 1$, an approximate kernel may display a pre-asymptotic Pattern (b)-like window governed by the exact-kernel effective theory. For $x \gg 1$, the kernel-lifted asymptotic behavior dominates and the joint peak also decays. The limits

$$\lim_{\Gamma \rightarrow \infty} \lim_{\varepsilon \rightarrow 0} R \quad \text{and} \quad \lim_{\varepsilon \rightarrow 0} \lim_{\Gamma \rightarrow \infty} R \quad (50)$$

need not coincide.

Remark 12.4 (Continuum caution). A finite matrix with a strictly positive Gaussian loss profile has no exact kernel. A continuum Gaussian that approaches zero at infinity may nevertheless have spectrum accumulating at zero. Finite-domain numerics alone do not decide the continuum, infinite-volume, and strong-dissipation limit order. These limits must be stated through Λ .

13 Admissible perturbations and destructive dephasing

Perturbations fall into three operational classes.

1. **Tangential or recalibratable perturbations:** they move $\mathcal{M}$ but preserve its rank and the protected cluster.
2. **Normal leakage perturbations:** they move the operating point away from $\mathcal{M}$ and are described by the finite- Γ tube.

3. **Kernel-destroying perturbations:** they lift the response-relevant protected cluster or add a decay larger than the Schur–Zeno linewidth.

For the current minimal realization, slow-coherence pure dephasing gives the model-specific numerical boundary

$$\gamma_{\phi e}^{\text{crit}} \simeq 0.41 \frac{J^2}{\Gamma_{\text{geo}}}. \quad (51)$$

The coefficient 0.41 is not universal. The structural lesson is that any fixed positive slow dephasing eventually dominates a linewidth proportional to J^2/Γ as $\Gamma \rightarrow \infty$. Therefore the codimension statement is made inside a declared admissible class, not over the entire physical parameter space.

14 Operational observability and certification diagnostics

The following diagnostics validate an application but are not additional axioms of RISEI.

Definition 14.1 (Möbius cancellation condition number). For a Möbius combination $\Omega_T = \sum_U \mu(U, T) \chi_U$, define schematically

$$C_{\text{cancel}} = \frac{\sum_U |\mu(U, T)| N_{\infty}[\chi_U]}{\max(N_{\infty}[\Omega_T], N_{\text{floor}})}. \quad (52)$$

A very large value indicates a numerically fragile subtraction and must trigger higher precision or analytic reconstruction.

A valid numerical or experimental claim must also declare:

- a signal-to-floor or signal-to-noise ratio;
- stability and GKSL physicality checks;
- fit-window and local-slope convergence;
- tangent/normal perturbation tests;
- cluster-resolved continuation rather than pole-label continuity;
- grid, domain, and precision convergence when applicable.

The ratio $N(\Omega_{12})/N(\chi_{\emptyset})$ is a descriptive amplitude diagnostic. In an indirect-readout realization it may be small for legitimate physical reasons and is therefore not, by itself, a pass/fail gate.

15 Finite intervention resources and accessibility

Definition 15.1 (Protocol cost).

$$\mathcal{C}[\pi] = \sum_j \left[a_j \int_{I_j} |\kappa_j(t)| dt + b_j \tau_j + c_j N_j \right]. \quad (53)$$

Definition 15.2 (Minimal identification cost). For a response distance D , target R_{target} , and tolerance ε ,

$$\mathcal{C}_{\text{req}}(\Gamma, \varepsilon) = \inf_{\pi} \{ \mathcal{C}[\pi] : D(R_{\pi}, R_{\text{target}}) \leq \varepsilon \}. \quad (54)$$

Definition 15.3 (Ideal cut as a derived limit). For a physical protocol family $\{\pi_a\}$,

$$R_S^{\text{ideal}} = \lim_{a \rightarrow \infty} R_{\pi_a}, \quad (55)$$

provided the limit exists and its dependence on the implementation family is declared.

16 Exact recovery of the earlier theories

Theorem 16.1 (Stationary linear embedding). *Assume a time-independent stable generator, a weak probe, first-order response, a static intervention, and a one-time mean readout. Then*

$$\chi_U(\omega) = c^\top (A_U - i\omega \mathbb{I})^{-1} p \quad (56)$$

up to Fourier/sign conventions. Thus the original stationary RISEI response is an exact section of the revised theory.

Corollary 16.2. *A fixed-frequency pointwise functional reduces ν_Φ to the original scalar exponent ν .*

The finite-time, multi-time, finite-resource, and ordered-intervention definitions of the first generalized theory remain unchanged. The present revision adds spectral-geometric data rather than replacing the protocol hierarchy.

17 Six-layer definition of revised generalized RISEI

The revised framework consists of six layers.

1. **Dynamics:** time-local CPTP evolution generated by $\mathcal{L}_{\Gamma, \pi, h}(t)$.
2. **Intervention protocol:** the ordered finite-resource object π .
3. **Response hierarchy:** $R_\pi^{(n)}$ for means, correlations, noise, and counting statistics.
4. **Irreducible decomposition:** $\Omega_T^{(n)}$ and $\Xi_{S,T}^{(n)}$.
5. **Protected spectral geometry:** Riesz clusters P_C , response relevance, the selection map F , the punctured set $\mathcal{M}$, and the admissible class $\mathcal{A}$.
6. **Asymptotic, limit, and resource classification:** ν_Φ , $\Sigma_\pi^{(n)}$, $k_{\min}^{(n, \Phi)}$, Λ , and $\mathcal{C}_{\text{req}}$.

The compact central object is

$$\mathfrak{R}_{\text{revised}} = \left\{ R_\pi^{(n)}, \Omega_T^{(n)}, \Xi_{S,T}^{(n)}, P_C, F, \mathcal{M}, \mathcal{A}, \nu_\Phi, \Sigma_\pi^{(n)}, \Lambda, \mathcal{C}_{\text{req}} \right\}. \quad (57)$$

RISEI is therefore a theory of how physical intervention protocols expose, suppress, or isolate internal response mechanisms, and of how the resulting visibility is controlled by protected spectral clusters, source/readout geometry, functional choice, finite resources, and limit order.

18 Current status of the central claims

Statement	Status	Support required or available
Möbius decomposition uniqueness	Exact	Finite Boolean-lattice inversion
Riesz cluster invariance	Exact	Contour functional calculus
Schur–Zeno expansion	Exact conditional	Semisimple kernel and bounded fast resolvent
No-kernel resolvent suppression	Exact conditional	Invertible response-relevant strong dissipator
Rank equals local codimension	Exact conditional	Constant-rank theorem
Pattern (b) on a punctured selection set	Conditional theorem schema	Requires norm hierarchy and nonzero observability coefficients

Statement	Status	Support required or available
Candidate A two reciprocal codimension-one branches	Numerical phenomenon plus model-level analytic geometry	Verified away from the intersection; intersection is a puncture
General N_1 power–log law	Conjecture / partial analytic target	Requires a uniform cluster-level bound
Finite- Γ tube $\propto \Gamma^{-1}$	Numerical phenomenon / conjecture	Broad model proof still required
Exact-to-approximate kernel crossover	Conjecture	Double-scaling and continuum-limit validation pending
Arbitrary-GKSL universality	Rejected	Counterexamples without response-relevant protected kernels
Universal codimension one	Rejected	Codimension is the real Jacobian rank
Tight universal onset predictor	Not established	Current predictor is informative but too conservative

19 Numerical certification protocol outside the core theory

A companion certification document should contain implementation-dependent gates rather than enlarging the basic theory. At minimum it should specify:

1. Riesz/cluster contour or continuation rules;
2. multiprecision and frequency-grid convergence;
3. N_∞ , N_1 , local valuations, and power–log model comparison;
4. C_{cancel} and numerical-floor thresholds;
5. Jacobian singular-value rank thresholds;
6. tangent, normal, off-manifold, and kernel-destroyed controls;
7. finite- Γ onset and right-censoring rules;
8. grid/domain/continuum extrapolation for field models;
9. explicit separation of theorem tests, model validation, and experimental detectability.

20 Deliberately deferred extensions

The revised core does not yet include unrestricted non-Markovianity, initial system–environment correlations, infinite-dimensional operator theorems, generic many-body scaling, or unrestricted nonlinear probe response. Continuum and many-body applications may be studied, but their additional limit structure must not be mistaken for an already proved part of the finite-dimensional theory.

21 Conclusion

The first generalized RISEI theory replaced a stationary scalar response by a finite-time, multi-time, finite-resource, ordered protocol hierarchy. The present revision completes the conceptual chain needed by the central phenomenon:

$$\boxed{\begin{aligned} \text{intervention} &\longrightarrow \text{protected Riesz cluster} \longrightarrow \text{source/readout selection geometry} \\ &\longrightarrow \text{functional hierarchy} \longrightarrow \text{operational signal.} \end{aligned}} \quad (58)$$

The theory does not claim that Pattern (b) occurs in every GKSL system. It predicts where such a hierarchy can occur, what spectral structure is required, how the observable-selection set is dimensioned, which perturbations preserve or destroy it, and why exact and approximate dissipative kernels belong to different asymptotic regimes.

A Notation

Symbol	Meaning
Γ	Dominant dissipative or asymptotic scale
S, T, U	Operational sectors or sector subsets
π	Ordered intervention protocol
$R_\pi^{(n)}$	Full-minus-protocol connected response
$\Omega_T^{(n)}$	Subset-Möbius irreducible response
$\Xi_{S,T}^{(n)}$	Order-irreducible response
Φ	Response functional
N_∞, N_1	Supremum and absolute-integral response norms
ν_Φ	Functional-dependent asymptotic valuation
$k_{\min}^{(n,\Phi)}$	Functional-dependent protection depth
$P_{\mathcal{C}}$	Riesz projector of a spectral cluster
P, Q	Protected and complementary spectral projectors
K_Z	Leading Schur–Zeno effective coupling
F	Real selection constraint map
$\mathcal{M}$	Punctured operational selection manifold
$\mathcal{P}$	Puncture or nonobservable subset
$\mathcal{A}$	Admissible perturbation class
Λ	Ordered limit protocol
C_{cancel}	Möbius cancellation condition number
$\mathcal{C}[\pi]$	Protocol resource cost
$\mathcal{C}_{\text{req}}$	Minimal identification cost

B Old-to-revised correspondence

Item	First generalized theory	Present revision
Primary response object	$R_\pi^{(n)}$	Unchanged
Irreducible structure	$\Omega_T^{(n)}, \Xi_{S,T}^{(n)}$	Unchanged; evaluated cluster-wise when needed

Item	First generalized theory	Present revision
Spectral object	Generator/resolvent used implicitly	Protected Riesz projector and invariant pole cluster
Robustness	Finite region, perturbation stability, or finite codimension	Declared admissible class; finite codimension is primary for selection manifolds
Observable dependence	Functional Φ and scaling signature	Adds source/readout selection map F and puncture set
Codimension	Not a central invariant	Real Jacobian rank of the active constraint map
Strong dissipation	General asymptotic scale	Explicit $\Gamma\mathcal{D} + \mathcal{B}$ Schur–Zeno reduction
Kernel condition	Not explicit	Response-relevant semisimple protected cluster is a central hypothesis
Finite- Γ behavior	Scaling and resource diagnostics	Adds onset, calibration tube, and right-censoring concepts
Limits	Primarily $\Gamma \rightarrow \infty$	Ordered limit protocol Λ and double scaling
Physical perturbations	General robustness checks	Recalibratable, normal leakage, and kernel-destroying classes
Pattern (b)	Candidate within functional hierarchy	Kernel-conditioned central phenomenon
Arbitrary-GKSL universality	Not explicitly excluded	Explicitly rejected